

Week 5: Sampling Distributions

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STA 220

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Overview

- Topics include:
 - Distribution of a Sample Mean
 - Distribution of a Sample Proportion
 - Mean and Variance of a Proportion
 - Sampling Distributions

Recall from last week: Random Variables

- Last week we introduced random variables, which are variables that take on specific realized values, at specific probabilities
- Random variables are denoted using capital letters
- Some random variables follow known distributions:
 - $X \sim \text{Bern}(p)$
 - $X \sim \text{Bin}(n, p)$
 - $X \sim N(\mu, \sigma^2)$

Recall from last week: Repeating Experiments

- Each time I can run an experiment, I can define my random variable X_i to represent the possible outcomes of the experiment
- If I decide I will run my experiment n times, then I have the random variables $X_1, X_2, \dots, X_n$ representing the possible outcomes for each time I run the experiment
- Each X_i has the same probability distribution representing the possible outcomes
- Oftentimes, we can use the random variables $X_1, X_2, \dots, X_n$ to represent our data

Goal for this week

- Functions of random variables are also random variables
- Our data from our experiment, $X_1, X_2, \dots, X_n$, are all random variables.
- This means that **sample statistics** (summary measures of the data such as sample mean; any function of the data) are also random variables
- Question: Can we find the distribution of the sample statistics?

Distribution of Sample Mean

Repeating an experiment

- Recall this example from last week.
 - Suppose I flip three coins, and I am interested in counting the number of heads I get
 - My random variable is $X = \# \text{ of heads}$, so $X \sim \text{Bin}(n = 3, p = 0.5)$
 - I repeat the experiment each day for 10 days
 - I will have my possible results for each day represented as $X_1, X_2, \dots, X_{10}$, and each of these follow $\text{Bin}(n = 3, p = 0.5)$
 - The mean of the results from all 10 days is:

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_{10}}{10}$$

- $\bar{X}$ is called the **sample mean**

Sample Mean

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_{10}}{10}$$

- Key Idea: $\bar{X}$ is a random variable, and thus has a probability distribution
- How is $\bar{X}$ a random variable?
 - Suppose I repeat the experiment for 10 days. I would get a value for $\bar{X}$ by averaging the results over the 10 days.
 - Suppose I repeat the experiment again for another 10 days. I would get another value for $\bar{X}$.
 - The value of $\bar{X}$ differs based on randomness.

Sample Mean

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_{10}}{10}$$

- How do we understand the distribution of $\bar{X}$?
 - Imagine if everyone in this class spent the next 10 days repeating this experiment 10 times (we all flip a coin 3 times daily, over the next 10 days).
 - We could all calculate $\bar{X}$, but likely our values would differ from one another
 - The distribution of all our $\bar{X}$'s represents the distribution of the sample mean $\bar{X}$

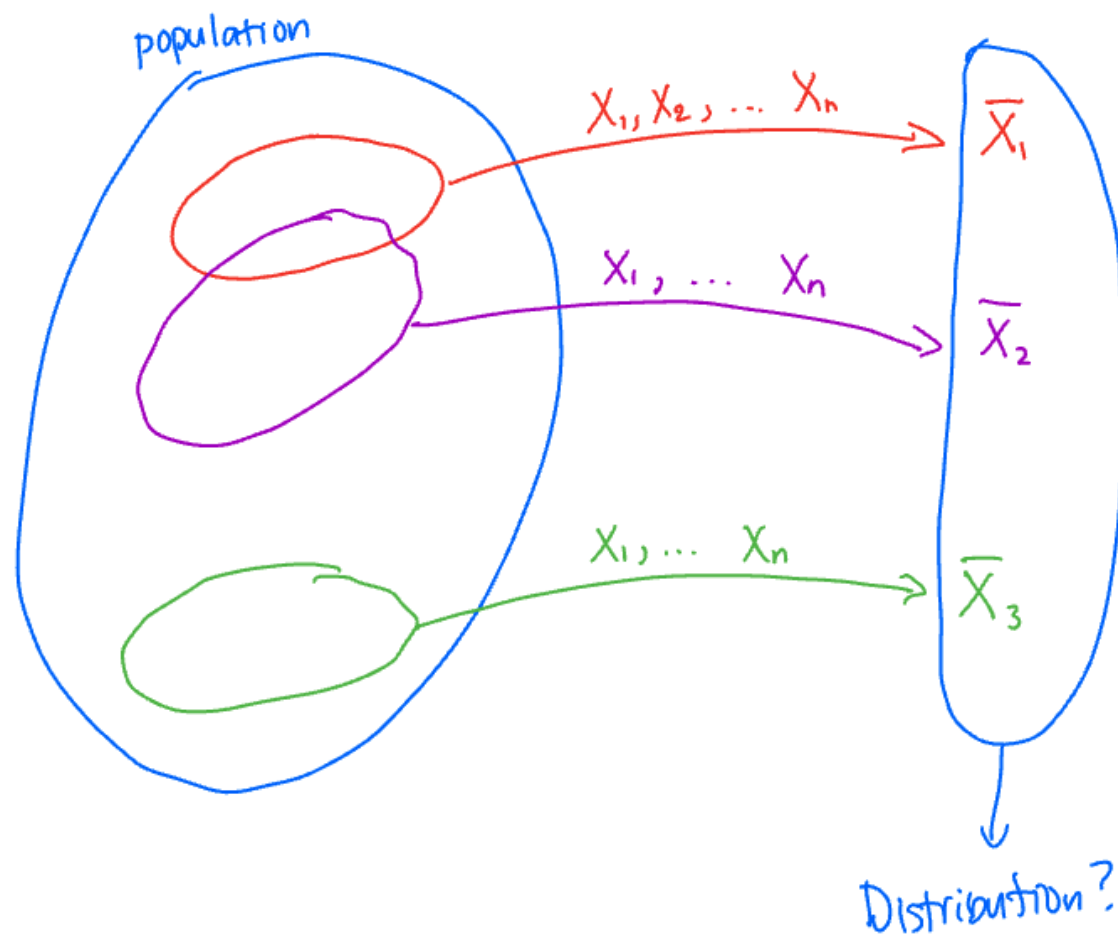
Another example of sample mean

- Example: Suppose we are interested in the average height of UofT students. We collect a sample of n measurements.
 - A measurement X (the height of a single student) is a random variable and follows some probability distribution
 - A sample mean $\bar{X}$ (the sample mean of a sample of size n) is also a random variable and follows some probability distribution
 - However, as we will see, they do not follow the same probability distribution
 - Key for now is to understand that these are 2 separate random variables and both have their own distributions

Sample Mean

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

- How do we understand the distribution of $\bar{X}$?
 - Imagine if we all went out and sampled n UofT students and recorded their heights
 - In other words, we each have a dataset $X_1, X_2, \dots, X_n$
 - We could all calculate $\bar{X}$ for the n people that we sampled, but likely our sample means would differ from one another
 - The distribution of all our $\bar{X}$'s represents the distribution of the sample mean $\bar{X}$



Distribution of Sample Means

- In summary, what does the *distribution of sample means* even mean?
 - The data I used to get the sample means were sampled from a distribution of possible values
 - If I were to keep sampling data forever, I would be able to eventually collect all the sample means corresponding to all possible samples
 - Then I would know how often I would see any particular sample mean
 - This is the distribution of sample means.
 - But, in real life, I don't ever see all of the possible sample means - just one of them!
 - We will see rules for how to determine the distribution of the sample mean, even though we only ever observe one of them.

Expected value and variance of $\bar{X}$

- Last week, we already saw some features of the distribution of $\bar{X}$
- $E(\bar{X}) = E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = E(X)$
- $Var(\bar{X}) = Var\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{Var(X)}{n}$
- This tells me that X and $\bar{X}$ have the same mean, but $\bar{X}$ has a smaller variance than X
 - This relates to the fact that $\bar{X}$ will give much more precise estimate than just X
- While this is a great start, we will now see how to obtain the full distribution of $\bar{X}$

LET'S *IMAGINE* THAT
WE WANT TO KNOW THE
AVERAGE VALUE IN A
CERTAIN POPULATION.

HOW MUCH
SODA DOES EACH
AMERICAN DRINK
PER DAY?



THEN LET'S IMAGINE WE GO OUT AND GATHER A WHOLE BUNCH OF
SEPARATE RANDOM SAMPLES FROM THAT POPULATION.

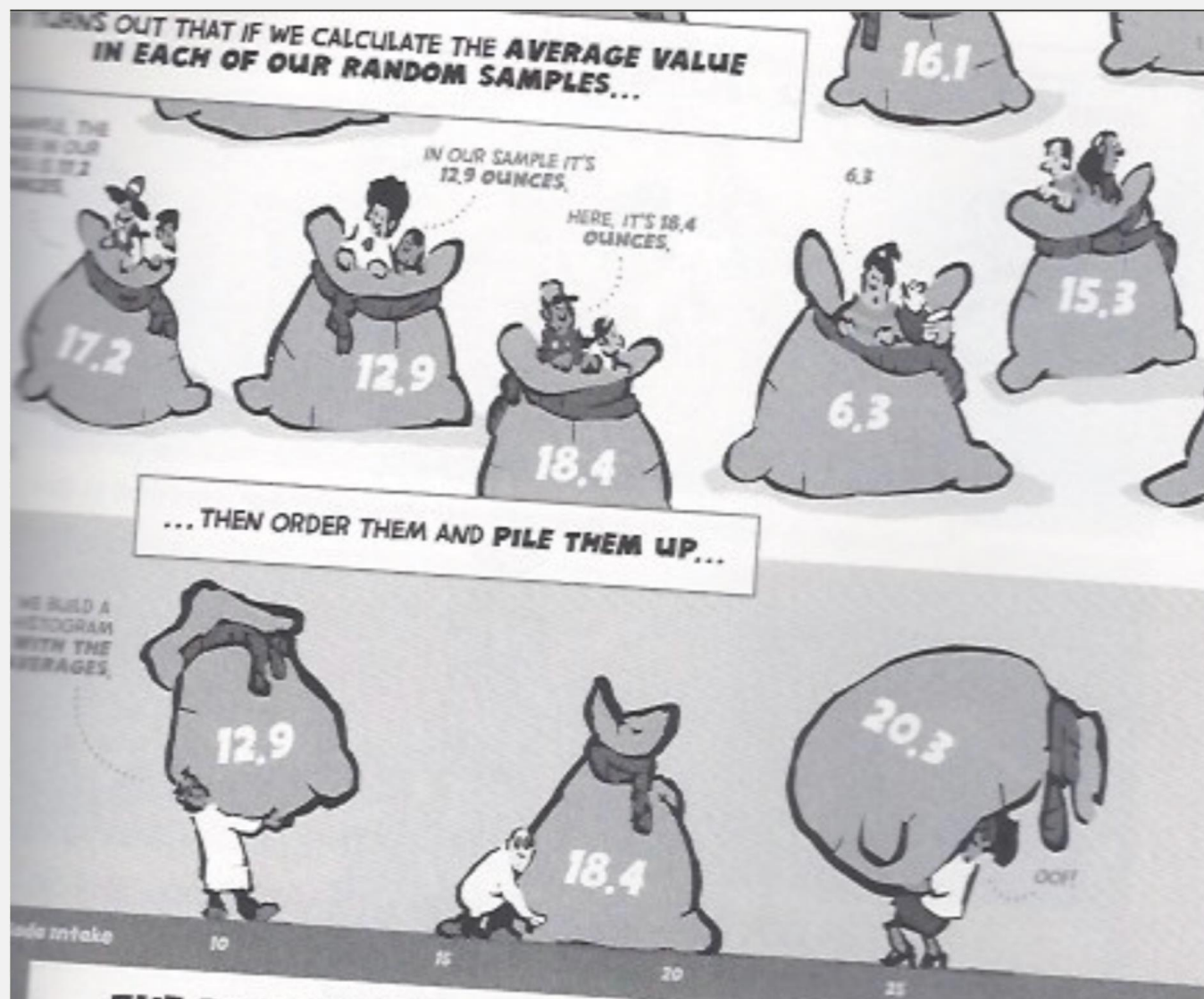
HERE ARE 50
RANDOM
AMERICANS,

HERE ARE
50 OTHER
RANDOM
AMERICANS,

HERE ARE
50 OTHER
RANDOM
AMERICANS,

EACH SAMPLE HAS
50 RANDOM
AMERICANS IN IT,

WE PUT EACH
SAMPLE IN A
BAG TO HELP US
KEEP TRACK OF
THEM,



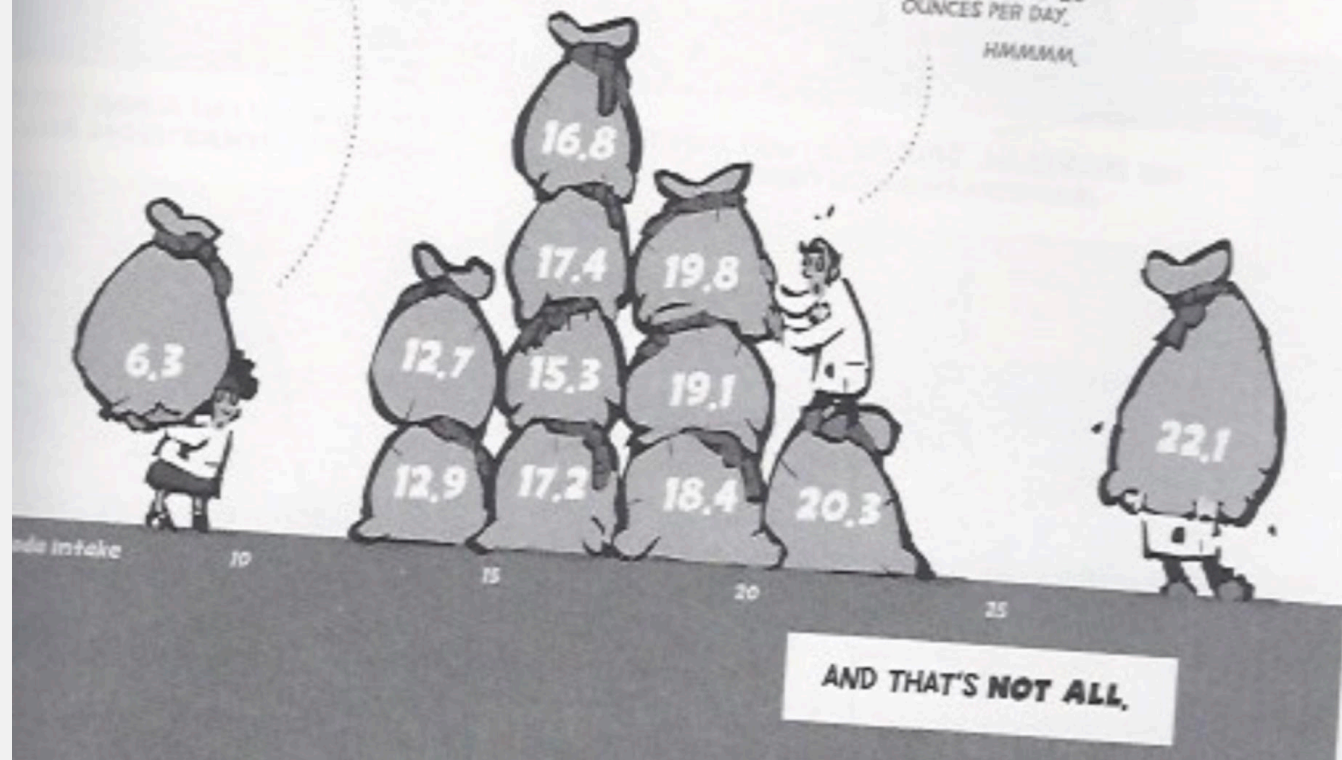
25

**... THE PILE OF AVERAGES WILL EVENTUALLY
START TO CLUMP TOGETHER!**

WE CAN EXPECT TO
SEE SOME EXTREME
AVERAGE VALUES
LIKE THIS ONE.

BUT MOST OF THE
AVERAGES CLUMP
AROUND HERE.

BETWEEN 15 AND 20
OUNCES PER DAY,
HMMMM.

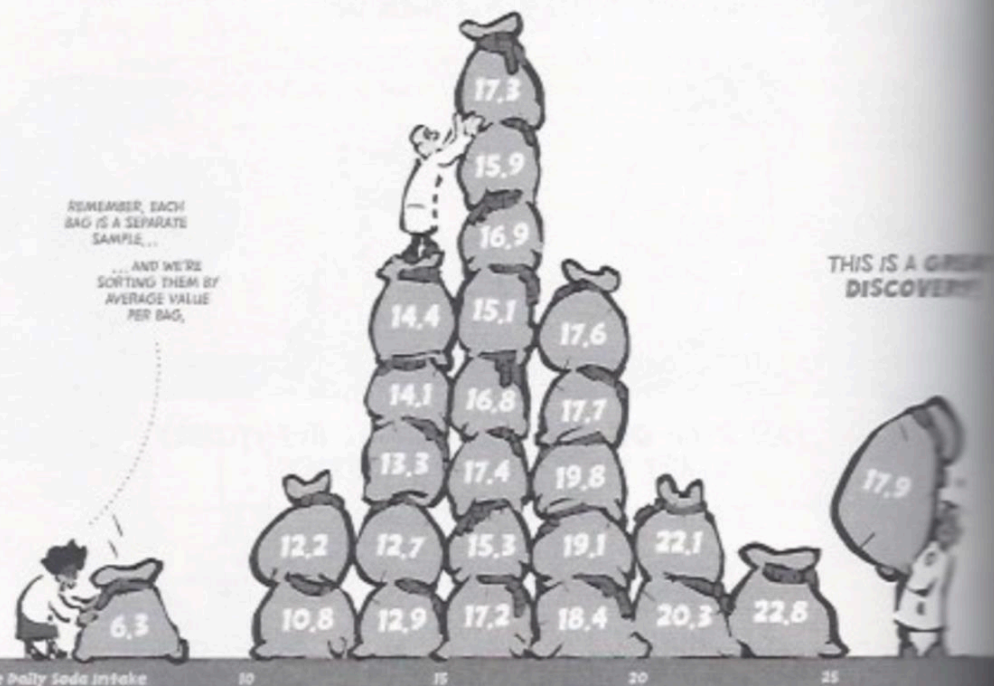


IT TURNS OUT THAT AS YOU PILE UP
MORE AND MORE SAMPLE AVERAGES...

BRING MORE!

WE WANT
GAZILLIONS!

... THE WHOLE PILE WILL TEND TO GET **MORE AND MORE
NORMAL-SHAPED.**



Description of the Illustrations

- Each bag has 50 Americans in it (random sample of 50 people)
- For each bag, I can compute the average soda consumption for those Americans
- Now each bag has a value attached to it (i.e. data)
 - it just happens to be a sample mean
- I can treat each value now as a single data point
- We know that when we collect data, we can examine the distribution... what I find is just that the sample means resemble a Normal distribution
 - The distribution of sample means is Normal (assuming some conditions are met)!

Sample Mean of Normal Observations

Sample Mean when $X_i \sim N(\mu, \sigma^2)$

Key findings when $X_i \sim N(\mu, \sigma^2)$

1. When measurements are random values that follow a normal distribution, the probability distribution of sample means is also a normal distribution.
2. The mean of the normal probability distribution of the sample means is the same as the mean of the probability distribution of the individual measurements.
3. The standard deviation of the probability distribution of the sample means is smaller than the standard deviation of the probability distribution of the individual observations. If there are n values in the random sample and σ is the standard deviation of the probability distribution of the individual observations, the standard deviation of the probability distribution of the sample means is $\frac{\sigma}{\sqrt{n}}$.

Let's take this apart step-by-step

1. When measurements are random values that follow a normal distribution, the probability distribution of sample means is also a normal distribution.

- Let $X_1, X_2, \dots, X_n$ be our random sample data, and $\bar{X}$ is the sample mean
- If $X_i \sim N(\mu, \sigma^2)$, then $\bar{X} \sim N(\mu, \sigma^2/n)$
- This is due to the fact that if you average independent normal variables, it is still normal
 - Related to this, if you sum up independent normal variables, it is still normal

Let's take this apart step-by-step

2. The mean of the normal probability distribution of the sample means is the same as the mean of the probability distribution of the individual measurements.

- Let $X_1, X_2, \dots, X_n$ be our random sample data, and $\bar{X}$ is the sample mean
- If $X_i \sim N(\mu, \sigma^2)$, then $\bar{X} \sim N(\mu, \quad)$
- This relates to the result that $E(\bar{X}) = E(X)$

Let's take this apart step-by-step

3. The standard deviation of the probability distribution of the sample means is smaller than the standard deviation of the probability distribution of the individual observations. If there are n values in the random sample and σ is the standard deviation of the probability distribution of the individual observations, the standard deviation of the probability distribution of the sample means is $\frac{\sigma}{\sqrt{n}}$

- Let $X_1, X_2, \dots, X_n$ be our random sample data, and $\bar{X}$ is the sample mean
- If $X_i \sim N(\mu, \sigma^2)$, then $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$
- This relates to the result that $Var(\bar{X}) = \frac{Var(X)}{n}$

IQ Example

IQ measurements are designed to follow a normal distribution with a mean of 100 and a standard deviation of 15. Suppose you collect a random sample 5 individuals and measure their IQ scores. State the probability distribution of the sample mean of these 5 measurements.

$$X_1, X_2, X_3, X_4, X_5 \sim N(100, 15^2)$$

$$\bar{X} \sim N\left(100, \frac{15^2}{5}\right)$$

Central Limit Theorem

Not normally distributed

- In the previous slides, we discussed a case when the individual observations were normally distributed
- But what if they are not normally distributed?
- This leads us into a different but similar set of results about the sample mean

Sample Mean (in General)

Key findings:

1. **Central Limit Theorem:** *For large enough sample size, the probability distribution of sample means is approximately a Normal distribution, regardless of the probability distribution of the individual measurements.*
2. The mean of the normal probability distribution of the sample means is the same as the mean of the probability distribution of the individual measurements.
3. The standard deviation of the probability distribution of the sample means is smaller than the standard deviation of the probability distribution of the individual observations. If there are n values in the random sample and σ is the standard deviation of the probability distribution of the individual observations, the standard deviation of the probability distribution of the sample means is $\frac{\sigma}{\sqrt{n}}$.

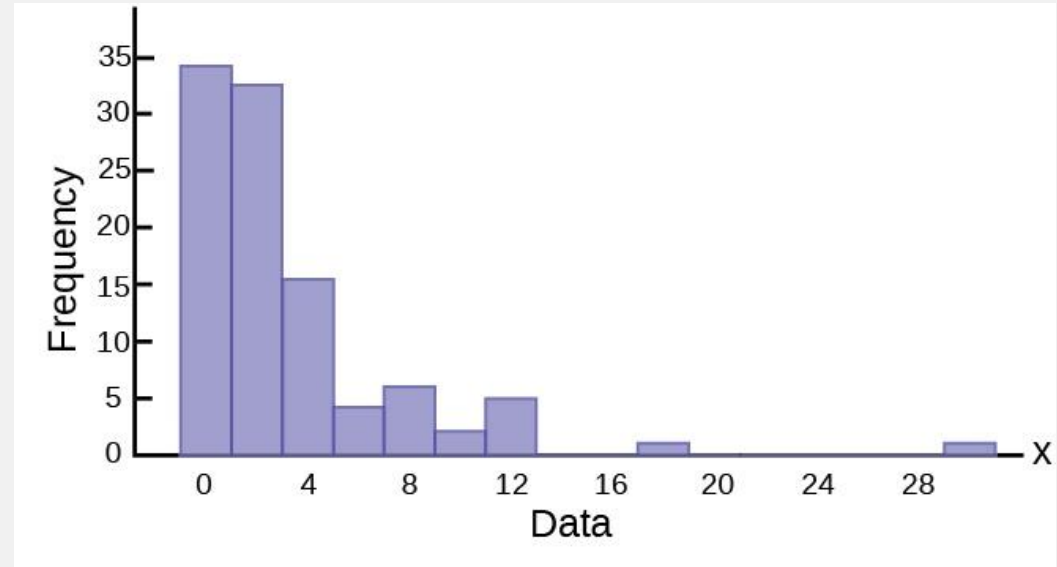
Note: The 2nd and 3rd points remain the same

Central Limit Theorem

- The central limit theorem asserts that: As more and more measurements are taken, the probability distribution for the possible averages of the measurements will converge to a Normal distribution
- Let $X_1, X_2, \dots, X_n$ be our random sample data, and $\bar{X}$ is the sample mean. Let $E(X_i) = \mu$ and $Var(X_i) = \sigma^2$
- CLT states that $\bar{X}$ is approximately $N(\mu, \frac{\sigma^2}{n})$ when n is sufficiently large
- The key is that the sample size n , must be large enough for the distribution of the sample mean to resemble a Normal distribution
- How large is large enough? It depends but $n \geq 30$ is a safe bet

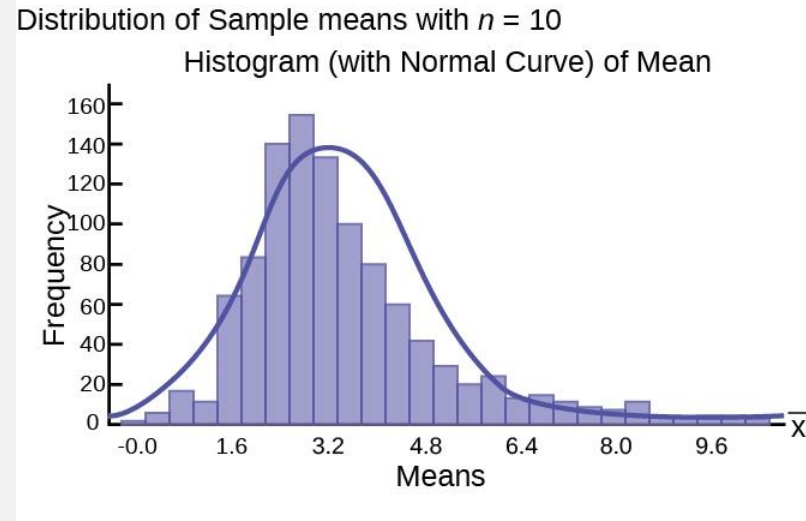
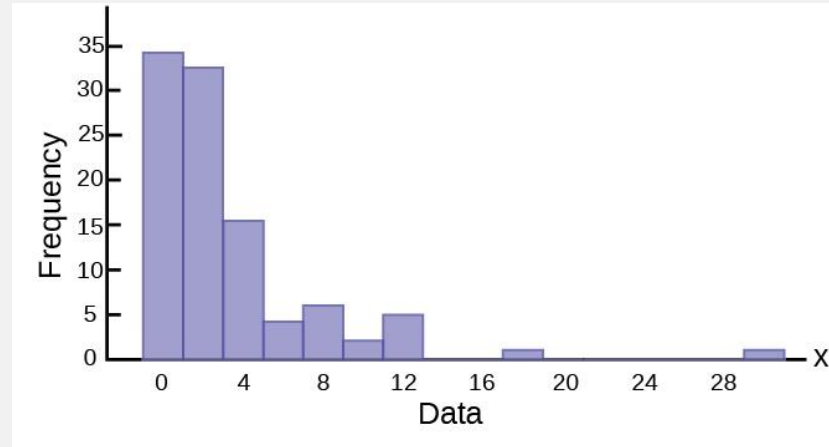
CLT Example

- Consider data on the time between subway arrivals.
- The data is not normal. Most come between 0-4 mins, but some are very delayed



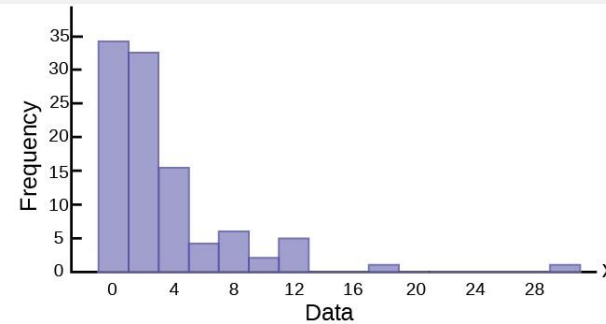
CLT Example

- Suppose we take 10 measurements and we have a sample of size 10. Then we calculate the sample mean.
- Let's repeat this 1000s of times... get a sample of size 10 and calculate the sample mean
- Now, let's examine the distribution of the sample mean by plotting all those sample means in a histogram

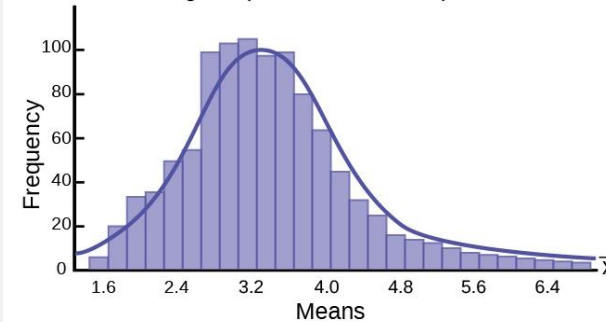


CLT Example

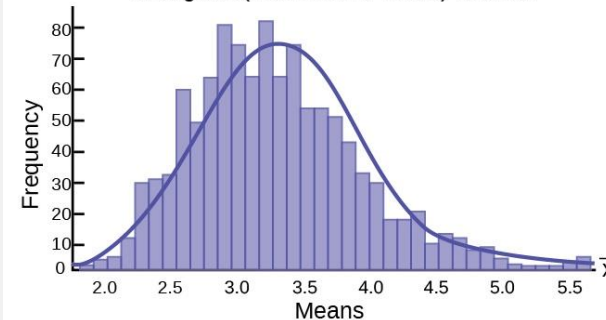
- We can do the same thing for a sample size of 25 and a sample size of 50
- The larger the sample size n , the more the distribution of $\bar{X}$ resembles a normal distribution
- Notice that this result holds true, even when the original measurements were not even close to a normal distribution



Distribution of Sample means with $n = 25$
Histogram (with Normal Curve) of Mean



Distribution of Sample means with $n = 50$
Histogram (with Normal Curve) of Mean



Example: Laptop lifespans

$$E(X) = 3.4$$
$$\text{Var}(X) = 0.8^2$$

The lifespan of a laptop has a mean 3.4 years with a standard deviation of 0.8 years. Suppose we collect information on a sample of 30 laptops.

a) What is the distribution for the average lifespan for this sample of 30 laptops?

Let X be the lifespan of a laptop. Then, $\bar{X}$ is the average lifespan of a sample of 30.

$$\bar{X} \sim N\left(3.4, \frac{0.8^2}{30}\right) \quad \text{by CLT since } n \geq 30$$

b) What is the probability that the average lifespan in the sample is more than 4 years?

$$P(\bar{X} > 4) = P\left(\frac{\bar{X} - 3.4}{0.8/\sqrt{30}} > \frac{4 - 3.4}{0.8/\sqrt{30}}\right) = P(Z > 4.1) \approx 0$$

Proportions

Proportions are Just Means

- Often when we are running experiments/conducting studies, we may not be interested in the sample average of continuous data.
- Sometimes we may be interested in data that consists of 1's and 0's (a Binomial experiment; repeating Bernoulli trials).
- So what would be the sample mean if we are dealing with Bernoulli data?
 - Example: If we are considering 5 independent Bernoulli trials and want the sample mean of the 1's and 0's, we could see

$$\frac{0 + 1 + 1 + 0 + 0}{5} = \frac{\text{Number of Successes}}{5} = \frac{2}{5}$$

- So my sample mean is $2/5$, which is just the **proportion** of 1's in 5 trials

Sample Proportion

- The number of successes is just the sum of these Bernoulli random variables: $X_1 + X_2 + \dots + X_n$
 - Taking the sum of 1's and 0's is the same as counting the number of 1's
- Due to this, sample mean of $X_1, X_2, \dots, X_n =$ the proportion of 1's
- Therefore, the **sample proportion $\hat{p}$** which is the proportion of successes in the sample, can be written as:

$$\hat{p} = \bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

- In other words, in the case where my data is 1's and 0's, the sample mean is now called **sample proportion $\hat{p}$**
- Goal: Find the distribution of the sample proportion $\hat{p}$

Example of Sample Proportion

- For example, suppose you had survey data on whether each randomly selected respondent wears glasses.
 - 1 = wears glasses
 - 0 = doesn't wear glasses
- The data would take the form of: 1, 0, 0, 1, 1, 0, ...
- The sample proportion $\hat{p}$ would be the proportion within the sample that wears glasses

Binomial Distribution when n increases

- We know that Bernoulli and Binomial random variables are discrete
- It turns out though that if we take larger and larger samples of Bernoulli random variables (i.e. look at a Binomial with bigger and bigger n) the distribution of the Binomial starts to look different
 - it starts to look less discrete in nature (smoother curve)
 - and it starts to look symmetric, bell-shaped and unimodal -> Normal
- So even though we have a discrete variable, the Central Limit Theorem still applies, as long as we are taking a large enough number of trials.
- This is because we saw that proportions are just sample means.

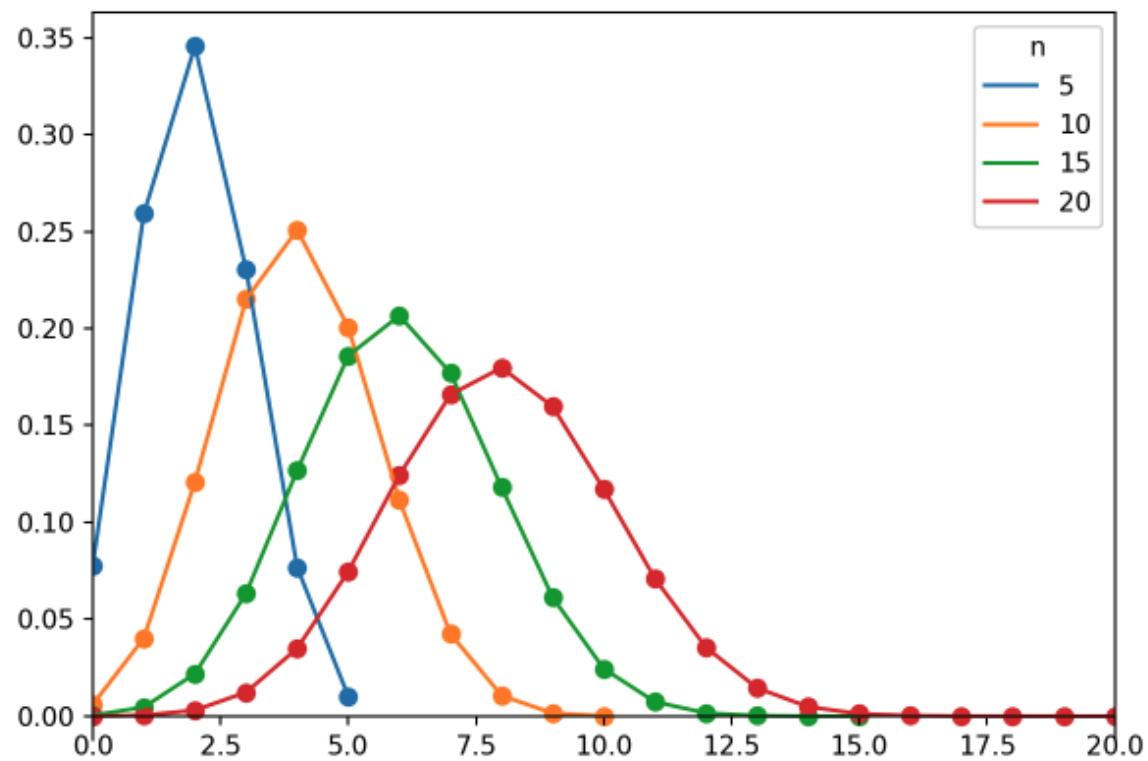


Figure 6.1: Probability mass functions for $\text{Binomial}(n, 0.4)$ distributions for $n = 5, 10, 15, 20$.

Bernoulli Mean

- In order to figure out what the *distribution of sample proportions* is, we need to know how to refer to the mean and variance of a Binomial random variable
- Recall that we can find the mean/expected value of a discrete variable by

$$E(X) = \sum_{i=1}^k x_i P(X = x_i)$$

- For a Bernoulli, we only have two probabilities, let's call $P(X = 1) = p$ and $P(X = 0) = 1 - p$. We can get the expected value for one Bernoulli by

$$E(X) = 0 \times (1 - p) + 1 \times p = p$$

Binomial Mean

- Now it's important to remember that a Binomial random variable is just the sum of independent Bernoulli random variables
 - Taking the sum of 1's and 0's is the same as counting the number of 1's
- So if X is Bernoulli, we found that $E(X) = p$. Then, we use one of the helpful properties of expected values,

$$E(X + Y) = E(X) + E(Y)$$

- Since a Binomial with n trials is just summing n Bernoulli's, we can find the mean of the Binomial by

$$E(Y) = E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n) = p + \cdots + p = np$$

$$Y \sim \text{Bin}(n, p)$$

Bernoulli & Binomial Variance

- We can basically do the same thing to get the variance of a Binomial distribution

- Start from the Bernoulli and use the definition of $\text{Var}(X)$:

$$\text{Var}(X) = \sum_i (x_i - E(X))^2 P(X = x_i) = (0 - p)^2(1 - p) + (1 - p)^2 p = p(1 - p)$$

- So since our Binomial is the sum of independent Bernoulli's, we can use another helpful property of variances:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

- We now get

$$\text{Var}(Y) = \text{Var}(X_1 + X_2 + \dots + X_n) = \text{Var}(X_1) + \dots + \text{Var}(X_n) = p(1 - p) + \dots + p(1 - p)$$

- So the variance of a Binomial distribution is $np(1 - p)$

Binomial Mean and Variance

- In summary, let $Y \sim \text{Bin}(n, p)$. Then,

$$E(Y) = np$$

$$\text{Var}(Y) = np(1 - p)$$

Back to the Sample Proportion

- Recall that the sample proportion is:

$$\hat{p} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

where $X_1, X_2, \dots, X_n$ are Bernoulli random variables

- Goal: Find the distribution of the sample proportion $\hat{p}$

Mean of $\hat{p}$

- Now we just use the same helpful property as finding the Binomial expected value

- We know that $\hat{p} = \frac{X_1 + X_2 + \dots + X_n}{n}$

- Also, each X_i is a Bernoulli random variable: $E(X_i) = p$

- If we want to find the expected value of $\hat{p}$, we can just do

$$E(\hat{p}) = \frac{E(X_1 + X_2 + \dots + X_n)}{n} = \frac{E(X_1) + E(X_2) + \dots + E(X_n)}{n} = \frac{\cancel{np}}{\cancel{n}} = p$$

- So, we get that $E(\hat{p}) = p$ where p is the probability of success for each X_i

Variance of $\hat{p}$

- Again we basically use the same logic as with the mean.
- We are still trying to find $Var(\hat{p})$ which is an average of independent Bernoulli variables
- We also have the useful result that when we have independent variables, $Var(X + Y) = Var(X) + Var(Y)$
- The variance of a Bernoulli is $p(1 - p)$ so we can use this to find the variance of $\hat{p}$

$$Var(\hat{p}) = Var\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{Var(X_1) + \dots + Var(X_n)}{n^2} = \frac{\cancel{np}(1 - p)}{\cancel{n^2}} = \frac{p(1 - p)}{n}$$

- So, we get that $Var(\hat{p}) = \frac{p(1-p)}{n}$

CLT for Proportions

- So putting all this together, we have that the Central Limit Theorem applies to proportions, as they are just sample means of Bernoulli variables.
- Therefore, as long as we are taking a large enough sample of Bernoulli measurements (n is large enough), proportion of successes $\hat{p}$ will be approximately Normal
 - Mean of the Normal distribution is $E(\hat{p}) = p$
 - Variance of the Normal distribution is $Var(\hat{p}) = \frac{p(1-p)}{n}$
- We can write $\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right)$ as the sampling distribution of the sample proportion

CLT for Proportions: Conditions

- The Central Limit Theorem for Proportions tells us that

$$\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right)$$

- This applies as long as the sample size is large enough
 - For proportions, we want the sample size, expected number of successes and expected number of failures to be large enough
 - Generally, we check that $n \geq 30$, $np \geq 10$, $n(1-p) \geq 10$

Example: Apples

When a truckload of thousands of apples arrives at a packing plant, a random sample of 150 apples are selected and examined for defects. Suppose that 8% of the apples on the truck have defects.

Find the distribution for the **number** of apples in the sample with defects.

Let Y be the number of apple in the sample with defects.

$$Y \sim \text{Bin}(n=150, p=0.08)$$

Find the distribution for the **proportion** of apples in the sample with defects.

$$\hat{p} \sim N\left(0.08, \frac{0.08 \times 0.92}{150}\right) \quad \text{by CLT} \quad \text{since} \quad \begin{aligned} n &\geq 30 \\ np &\geq 10 \\ n(1-p) &\geq 10 \end{aligned}$$

Example: Apples

The whole truckload of apples will be rejected if more than 5% of the sample have defects. What is the probability that the truckload will be rejected?

$$P(\hat{p} > 0.05) = P\left(\frac{\hat{p} - 0.08}{c} > \frac{0.05 - 0.08}{c}\right)$$

$$= P(Z > -1.35)$$

$$= 0.91$$

$$\text{Let } c = \sqrt{\frac{0.08 \times 0.92}{150}}$$

Overview of Sampling Distributions

Sampling Distributions

$$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$$

$$\hat{p} \sim N(p, \frac{p(1-p)}{n})$$

- A **sample statistic** is any function of your data $X_1, X_2, \dots, X_n$. The sample mean is only one example of a sample statistic
- **Sampling distributions** refer to the probability distribution of a sample statistic (such as sample mean, sample proportion, etc.).
 - The probability distribution accounts for the variability from one sample to another
- The sampling distributions will be in terms of **parameters**. Parameters describe the population (such as the mean in the whole population, μ , or the proportion in the whole population, p)
 - Parameters are fixed values. They are not random variables

Sampling Distributions

- In summary, we have seen two sampling distributions
 - Sample mean: $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$
 - Sample proportion: $\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right)$
- These sampling distributions are very important for making inference about certain aspects of a population where it is not possible to measure everyone.
 - Inference is just a fancy word for guessing the value of something, but it also incorporates the idea that there is error and randomness associated with that guess

Sampling Distributions

- Let's take our defective apple example again.
- For the purpose of actually being able to calculate things on a test, you are given the **true proportion** of apples in the truckload that have defects ($p = 0.08$).
 - In real life, you don't know this value.
 - In fact, you would be using your sample of 150 apples to **estimate** the value of p that you don't know
 - So we say that $\hat{p}$ is an estimate for p and we use the sampling distribution to talk about how close this estimate really is to the unknown true proportion.
 - The sampling distribution is therefore the probability distribution of all possible values of the estimator $\hat{p}$
- The variance of the sampling distribution represents the variability of my estimator due to the fact that I took a sample.

Exercise

The weight of potato chips in a medium-size bag is stated to be 300g. The amount that the packaging machine puts in the bags is believed to have a Normal model with mean 306g and standard deviation 3.6g.

a) We want to determine how often the machine is overfilling the bags, so we take a sample of 100 bags. What is the probability that the average of the sampled bags has weight more than 305g?

Let X be the mass of a bag of chips.

$$X \sim N(306, 3.6^2) \quad n=100$$

$$\bar{X} \sim N\left(306, \frac{3.6^2}{100}\right)$$

$$\begin{aligned} P(\bar{X} > 305) &= P\left(\frac{\bar{X} - 306}{3.6/10} > \frac{305 - 306}{3.6/10}\right) = P(Z > -2.78) \\ &= 0.9973 \end{aligned}$$

Exercise Continued

b) Based on the previous exercise: Some of the bags of chips are sold in packs of 5. What is the expected number of bags in a pack that have been filled with more than 305g of chips? (Assume the bags chosen to be in a pack are selected at random)

What is the prob. that a single bag of chips has more than 305g?

$$P(X > 305) = P\left(\frac{X - 306}{3.6} > \frac{305 - 306}{3.6}\right) = P(Z > -0.278) \\ = 0.6103$$

Let Y be the number of bags in a 5-pack with more than 305g.

$$Y \sim \text{Bin}(n=5, p=0.6103)$$

$$E(Y) = 5 \times 0.6103 = 3.0515$$

Exercise Continued

c) What is the distribution for the proportion of bags with weight more than 305g in a sample of 100 bags?

Let $\hat{p}$ be the proportion of bags (out of 100) that have more than 305g.

$$\text{So, } \hat{p} = (X_1 + X_2 + \dots + X_{100}) / 100$$

$$\hat{p} \sim N\left(0.6103, \frac{0.6103(1 - 0.6103)}{100}\right)$$